

Ishay Shaul

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Current Computer Science and Statistics double-major with a focus on **computational efficiency and systems logic**. Extensive experience implementing high-performance solutions, ranging from second-order optimization models in Python to complex multiplayer game systems. Proficient in **C++, Java, and Python**, with a strong academic and practical foundation in **Data Structures and Algorithms**. Focused on the intersection of mathematical theory and robust system design, with a proven ability to develop secure database architectures and efficient, object-oriented software.

Projects:

- **Text Summarizer(Python, Javascript, HTML, CSS):** Develop a complete codebase that integrates the **backend with the frontend**, while gaining practical experience working with Hugging Face Transformers.
 - **Multi user authentication system(Java, SQL):** Built a multi-user authentication platform utilizing BCrypt encryption and SQLite. Features include a custom password strength validator (Regex), automated brute-force protection (lockout logic), and a secure password generation engine. Project focused on **Cryptography** and **Databases**.
 - **Mathematical Optimization & Gradient Descent (Python):** Developed a project evaluating first-order and second-order gradient descent methods, implementing an encoder-decoder while focusing on convergence rates and computational efficiency.
 - **One-Person "Splitwise" Application(C++):** Engineered a personal finance management tool to track and balance expenses. Focuses on user interface and data structures.
 - **Chinese Checkers(Python):** A multiplayer game of chinese checkers, focusing on Interface between players and logic for the Computer character.
 - **Chrome Dino like Game(Java):** Designed and implemented a chrome Dino (the game in chrome) like game, focusing on object oriented programming design patterns.
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Education:

B.S. in Computer Science

Hebrew University of Jerusalem | expected to graduate in August 2027

- Notable courses: Introduction to Computer Science, C/C++ for practical Algorithms,
- Data Structures, Algorithms, OOP, Intro to Machine Learning, Optimization, Regression and linear models for Computer Science, Intro to statistics, Statistics lab, nand2tetris(equivalent to Computer Architecture).

Himmelfarb High School | 2015 - 2018

- finished with 114 GPA
 - Majored in Physics and History, alongside Honors class in Mathematics.
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Military Service: Mar 2020 - Nov 2022

Golani Brigade:

- Full service as a private and sergeant.
 - Leading and commanding teams of up to two dozen subordinates.
 - Decision making under pressure.
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Skills:

- **Specializations:** Software Development, Statistical Modeling, and AI/Machine Learning concepts.
 - **Languages:** Fluent in English and Hebrew.
 - **ML/Data Libraries:** NumPy, Pandas, and PyTorch.
 - **Programming Languages:** Python, C, Java, C++, R.
 - **Operating Systems & Environments:** Linux/WSL, Git, VSCode, Clion.
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Professional Qualities

- **Leadership:** Experience in leadership, command, and leading teams in the field.
- **Communication:** Excellent interpersonal communication and experience working with clients.
- **Work Ethic:** High work ethic and the ability to work effectively in a team.
- **Reliability:** Strong decision-making skills under pressure.